

Department of Computer and Software Engineering**SE: Machine Learning****Course Instructor:****Date:****Day:****Semester:****Lab 7: Agglomerative Clustering and Dendrograms**

Lab Title	CLO	BT	PLO	Weightage
Agglomerative Clustering and Dendrograms				

Name	Roll Number	Report /10	Viva /5	Total /15

Checked on: _____

Signature: _____

1 Learning Outcomes

After completing this lab, students will be able to:

- Understand hierarchical clustering.
- Implement agglomerative clustering from scratch.
- Compare different linkage methods.
- Interpret dendrograms.

2 Dataset

We will use a synthetic dataset generated using:

```
sklearn.datasets.make_blobs
```

Dataset properties:

- Number of samples: 100
- Number of clusters: 3
- Features: 2 (for visualization)

3 Agglomerative Clustering

Agglomerative clustering is a bottom-up hierarchical clustering method.

Algorithm:

1. Start with each point as its own cluster.
2. Find the closest pair of clusters.
3. Merge them into a single cluster.
4. Repeat until desired number of clusters is reached.

4 Linkage Methods

Different linkage methods define cluster distance differently.

4.1 Single Linkage

$$d(C_1, C_2) = \min ||x - y||$$

Produces chain-like clusters.

4.2 Complete Linkage

$$d(C_1, C_2) = \max ||x - y||$$

Produces compact clusters.

4.3 Average Linkage

$$d(C_1, C_2) = \text{mean}||x - y||$$

Balanced clustering approach.

5 Part A: Distance Matrix (3 Marks)

Implement:

```
compute_distance_matrix(X)
```

Requirements:

- Compute pairwise Euclidean distances
- Output should be an NxN matrix
- Matrix must be symmetric
- Diagonal values must be zero

6 Part B: Linkage Distance (4 Marks)

Implement:

```
linkage_distance(cluster1, cluster2, X, method)
```

Support the following methods:

- Single linkage
- Complete linkage
- Average linkage

7 Part C: Find Closest Clusters (3 Marks)

Implement:

```
find_closest_clusters(clusters, X, method)
```

Steps:

- Compute distance between all cluster pairs
- Return indices of closest clusters
- Return the corresponding distance

8 Part D: Agglomerative Clustering (5 Marks)

Implement:

```
fit(X)
```

Steps:

1. Initialize each point as its own cluster
2. Iteratively merge closest clusters
3. Stop when desired number of clusters is reached

Store:

- Cluster labels
- Merge history (for dendrogram)

9 Part E: Dendrogram (Bonus)

Implement:

```
plot_dendrogram(history)
```

Steps:

- Convert merge history to linkage format
- Plot dendrogram using scipy

10 Part F: Comparison with sklearn (Bonus)

Use:

```
sklearn.cluster.AgglomerativeClustering
```

Compare:

- Cluster assignments
- Effect of linkage methods

11 Expected Output

Your program should:

- Perform agglomerative clustering
- Display cluster plots
- Show differences between linkage methods
- (Bonus) Display dendrogram